

Spatial Data Analysis Portfolio

by

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Session 5

Thinking London Spatially Further (Spatial Query with SpatiaLite Advance)

Task 1: The query of london_guinness_book.

Result:

London Borough	Region	Area (sq.Km)	Density: Population / sq.Km	North	South	East	West
City	Inner	3.14938	2413.173	182206.1	180406.7	533839.6	530967.7
Islington	Inner	14.85665	14202.39	187968.7	181659.1	533484.5	528832.4
Enfield	Outer	82.20019	3860.088	200933.9	191460.9	537813	525568.6
Croydon	Outer	86.49557	4264.959	171048.7	155850.8	539657.9	528185.7
Havering	Outer	114.4573	2094.231	194889.3	178410.9	561957.5	548106.9
Hillingdon	Outer	115.7011	2435.586	193619.8	173846.2	512764.7	503568.2
Bromley	Outer	150.1349	2091.453	173658.3	156480.9	550541.2	533531.5

Description

In this task I used spatial sql function for the analysis of London boroughs and find the boroughs with the most extreme values northmost, southmost or having least and most density population etc.

Spatial SQL can be used to explore urban characteristics. Spatial data helps to reveal different patterns which can be both demographic and geographic.

Task 4: The count of underground station by Borough

Result

London Borough Name	Number of Tube Stations
Camden	19
City	14
Hackney	1
Hammersmith and Fulham	15
Haringey	6
Islington	11
Kensington and Chelsea	11
Lambeth	10
Lewisham	2
Newham	27
Southwark	5
Tower Hamlets	27
Wandsworth	6
Westminster	36
Barking and Dagenham	5
Barnet	12
Brent	21
Ealing	15
Enfield	4
Greenwich	5
Harrow	10
Havering	4
Hillingdon	15
Hounslow	9
Merton	5
Redbridge	10
Richmond upon Thames	2
Waltham Forest	4

Description

In this task the number of underground tube stations in each London borough are counted and the results suggest that Westminster is the borough with highest number of tube station and boroughs such as Hackney, Lewisham and Richmond upon Thames have very less tube stations in London.

Task 6: Motorways across London (Intersection)

Result:

Motorway	London Borough Name	Region
M1	Barnet	Outer
M1	Harrow	Outer
M11	Redbridge	Outer
M25	Bromley	Outer
M25	Enfield	Outer
M25	Havering	Outer
M25	Hillingdon	Outer
M4	Hillingdon	Outer
M4	Hounslow	Outer

Description

The above table demonstrates the intersection of motorways across different London boroughs. One of the key observations is that all these motorways are located in outer region. Thus, it can be concluded from the results that major motorways are mostly located in outer parts of the city.

Applications of these commands include urban and transport route planning along with the traffic control and management.

Task 7: Underground stations within 1 km of motorways

Result:

Motorway	Station	Line	distance to motorway < 1km
M4	Boston Manor Underground Station	Piccadilly	1
M11	Chigwell	Central Line	1
M4	Chiswick Park Underground Station	District	1
M1	Colindale Underground Station	Northern	1
M11	Debden	Central Line	1
M4	Gunnersbury Underground Station	District	1
M1	Hendon Central Underground Station	Northern	1
M11	Roding Valley	Central Line	1
M11	South Woodford	Central Line	1
M25	Terminal 5	Piccadilly	1
M11	Theydon Bois	Central Line	1

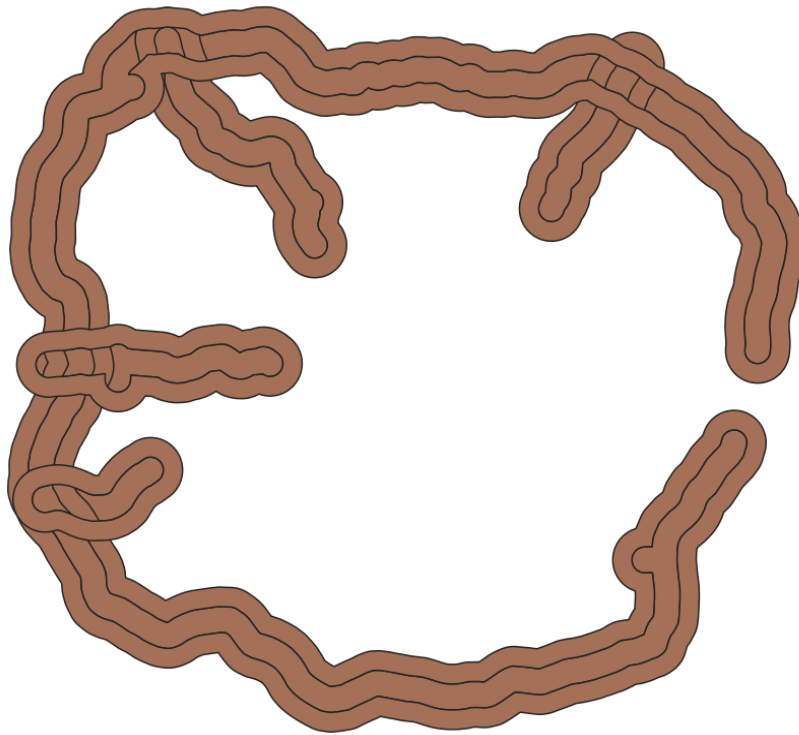
Description

The above table depicts the underground stations that are located within the distance of 1 kilometer of major motorways around London. It is clear from the data that more stations are near to M11.

Applications of these commands include urban and transport route planning along with the traffic control and management.

Task 8: Export maps (in PNG format from SpatiaLite, or in other formats from QGIS)

Result:



Description

This task shows how buffer zones can be generated around linear features, in this case motorways, and how results can be stored and represented. The 1 km buffer zones are represented as areas around the motorways. The second ring includes the area from 1 to 2.5 km and it forms the outer circle that separates the near and far areas.

Such maps can be crucial to get an idea of which areas might be affected by the motorways and thus it can be beneficial

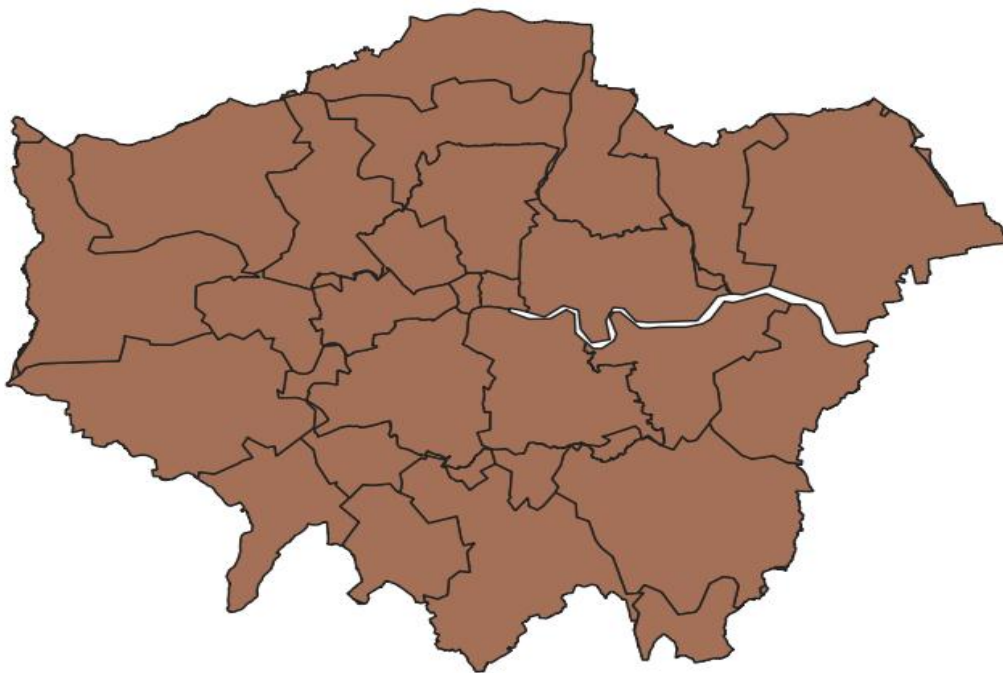
These commands can be very useful in urban and emergency planning along with environmental impact assessment

Session 6

Preparation / Integration of Spatial Data (Spatial Operation and Transformation with QGIS)

Task 2: The map of Intersect (AND) from Task 2

Result:



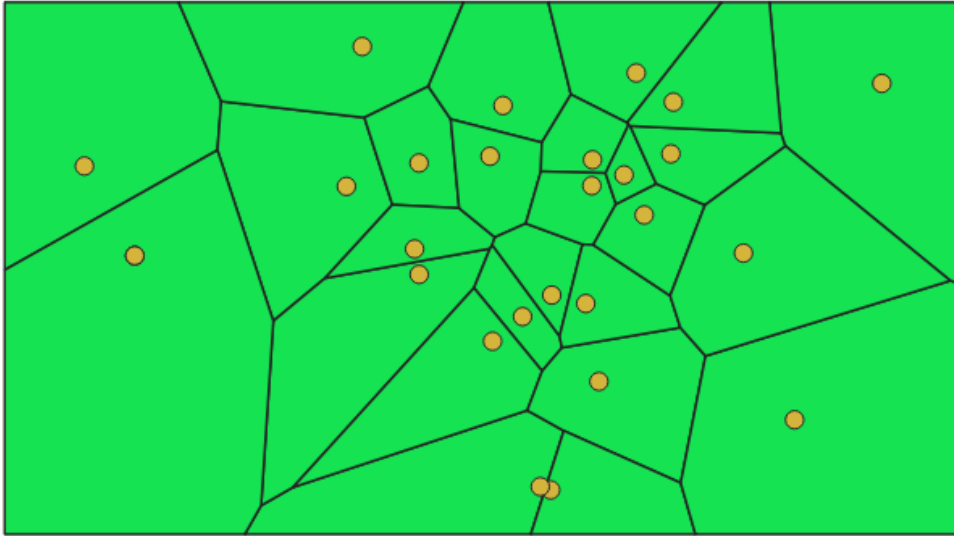
Description

In this task, with the use of intersect tool (Vector, Geoprocessing Tools , intersect (AND)) postcode area map (postarea.shp) is combined with Inner and Outer London boundary map (london_io.shp).

The result demonstrates how areas with larger postcode are broken down into smaller parts depending on their location, whether they are in inner, outer or outside London.

Task 3: Voronoi Polygon

Result:



Description

In this task Voronoi Polygons tool was used in order to create catchment areas for each credit union (Vector, Geometry tools , Voronoi Polygons) with 10 % buffer region.

From the results it can be concluded that the map is divided into zones and each credit union is surrounded by a zone.

Task 4: Height image from Task 4

Result:



Description

In this task, to extract the height of features from LiDAR data, Digital Surface Model (DSM) and Digital Terrain Model (DTM) data of the Ravenscourt Park area in Hammersmith, London were used.

In the first step two raster TQ2278_DTM_2M.asc which represents ground elevation and TQ2278_DSM_2M.asc which represents elevation including buildings, trees, etc were loaded.

In the second step, Raster Calculator was used and the calculations were made using formula "DSM layer" - "DTM layer". Resulting image is the map of Buildings around Ravenscourt Park, Hammersmith, London.

Task 6: Save the vector layer of London_boroughs as CSV

Result:

NAME	HECTARES	LA_CODE01	LA_CODE11	IN_OUT	X_CENTROID	Y_CENTROID	Pop	HH	Pop_den	HH_size
Kingston upon Thames	3726.117	00AX	E09000021	Outer	519505	166134	163900	63639	43.987	2.575
Croydon	8649.554	00AH	E09000008	Outer	533922	163449	368900	145010	42.65	2.544
Bromley	15013.49	00AF	E09000006	Outer	542035	165071	314000	130862	20.915	2.399
Hounslow	5658.541	00AT	E09000018	Outer	514531	174962	259100	94902	45.789	2.73
Ealing	5554.428	00AJ	E09000009	Outer	515796	181961	340700	124082	61.338	2.746
Havering	11445.74	00AR	E09000016	Outer	555032	186650	239700	97199	20.942	2.466
Hillingdon	11570.12	00AS	E09000017	Outer	508167	183733	281800	100214	24.356	2.812
Harrow	5046.275	00AQ	E09000015	Outer	515356	189957	242400	84268	48.035	2.877
Brent	4323.27	00AE	E09000005	Outer	520507	186468	314700	110286	72.792	2.853
Barnet	8674.837	00AC	E09000003	Outer	523472	191948	364000	135916	41.96	2.678
Lambeth	2724.94	00AY	E09000022	Inner	531118	175167	310200	130017	113.837	2.386
Southwark	2991.34	00BE	E09000028	Inner	534034	175751	293500	120422	98.117	2.437
Lewisham	3531.706	00AZ	E09000023	Inner	537888	174589	281600	116091	79.735	2.426
Greenwich	5044.19	00AL	E09000011	Outer	542377	176482	260100	101045	51.564	2.574
Bexley	6428.649	00AD	E09000004	Outer	549409	175841	234300	92604	36.446	2.53
Enfield	8220.025	00AK	E09000010	Outer	531691	196197	317300	119916	38.601	2.646
Waltham Forest	3880.793	00BH	E09000031	Outer	537327	190698	262600	96861	67.667	2.711
Redbridge	5644.225	00BC	E09000026	Outer	544061	189480	284600	99105	50.423	2.872
Sutton	4384.698	00BF	E09000029	Outer	526735	163640	193600	78174	44.154	2.477
Richmond upon Thames	5876.111	00BD	E09000027	Outer	517721	172687	189100	79835	32.181	2.369
Merton	3762.466	00BA	E09000024	Outer	526025	169505	202200	78757	53.741	2.567
Wandsworth	3522.022	00BJ	E09000032	Inner	525627	174183	308300	130493	87.535	2.363
Hammersmith and Fulham	1715.409	00AN	E09000013	Inner	523930	179217	179900	80590	104.873	2.232
Kensington and Chelsea	1238.379	00AW	E09000020	Inner	525764	179843	155900	78536	125.89	1.985
Westminster	2203.005	00BK	E09000033	Inner	527508	180816	223900	105772	101.634	2.117
Camden	2178.932	00AG	E09000007	Inner	527755	184282	225000	97534	103.262	2.307
Tower Hamlets	2157.501	00BG	E09000030	Inner	536458	181372	263000	101257	121.9	2.597
Islington	1485.664	00AU	E09000019	Inner	531158	184815	211000	93556	142.024	2.255
Hackney	1904.902	00AM	E09000012	Inner	534562	185084	252100	101690	132.343	2.479
Haringey	2959.837	00AP	E09000014	Inner	531279	189350	258900	101955	87.471	2.539
Newham	3857.806	00BB	E09000025	Inner	541567	183188	314100	101519	81.419	3.094
Barking and Dagenham	3779.934	00AB	E09000002	Outer	547682	186108	190600	69681	50.424	2.735
City	314.942	00AA	E09000001	Inner	532404	181306	7600	4385	24.131	1.733

Description

In this task, two new variables population density and household size were created and calculated to the London borough's shapefile with the help of QGIS attribute table tools.

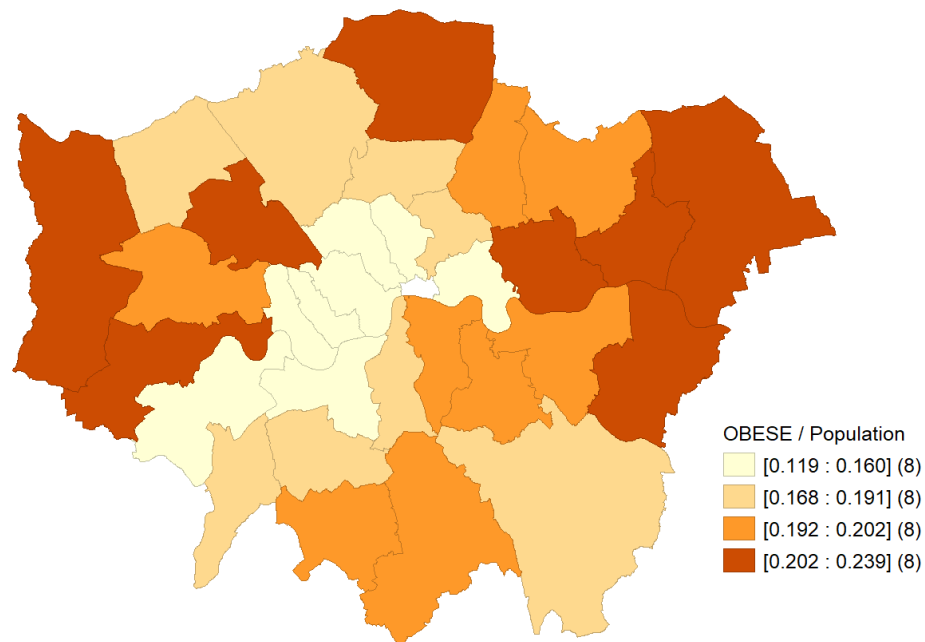
The results suggest that Inner London is more densely populated as Boroughs like Islington (142.0), Hackney (132.3), and Kensington and Chelsea (125.9) have the highest population densities. Household sizes don't vary on a large scale and Newham (3.09) has the highest household size.

Session 7

Exploring and Mapping Social Landscape of London (Basic Geostatistics and Thematic Mapping with QGIS and GeoDa)

Task 1: Export rate and risk maps of obesity created in GeoDa

Result (Raw Rate):

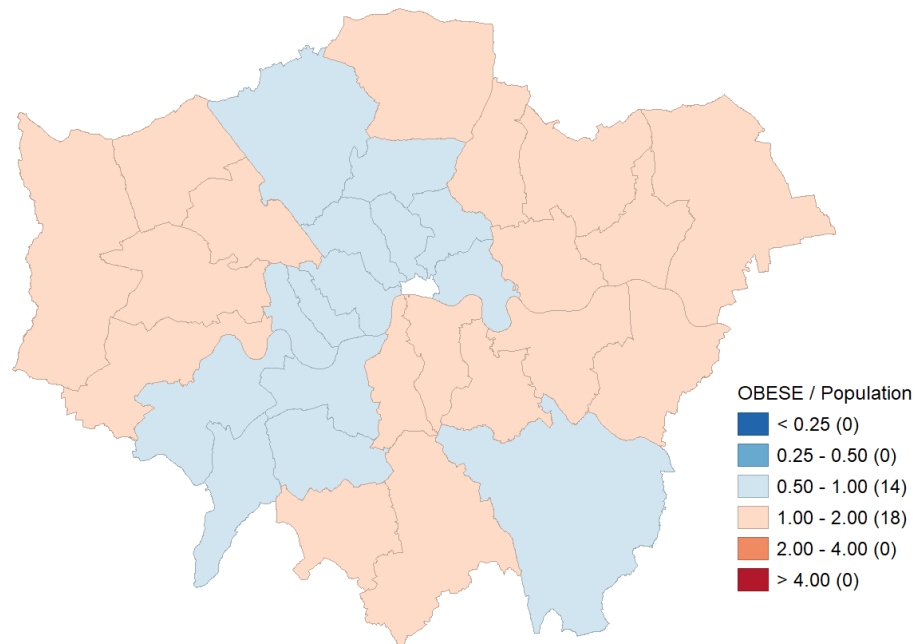


Description

The above raw rate map depicts spatial distribution of risk of obesity across different boroughs of London. This map reveals that the risk of obesity in outer areas is higher.

Such mapping can help officials in the public health sector to spread awareness across the targeted high-risk areas. The mapping process could further be improved by including the socioeconomic factors and by making the map interactive and easy to interpret.

Result (Excess Risk):

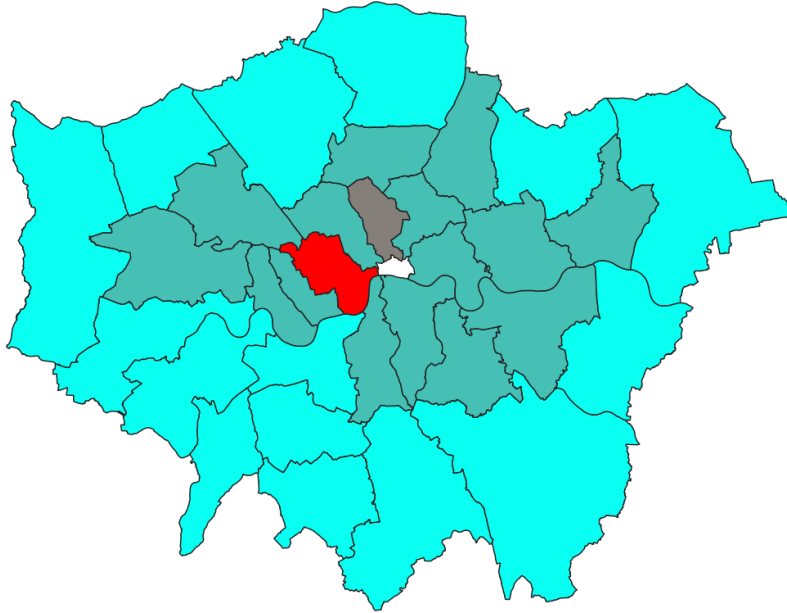


Description

The above map is an excess risk map that depicts the spatial difference in the risk of obesity across different boroughs of London. It suggests that in majority of the areas, the excess risk is moderate. Such analysis can help in planning public health as it helps to identify the areas or boroughs that need more attention.

Task 2: Export thematic map of crime created in QGIS from

Result:



Description

The above map depicts the crime rates (CRIME_1000) across different boroughs of London with the use of graduated symbology in QGIS.

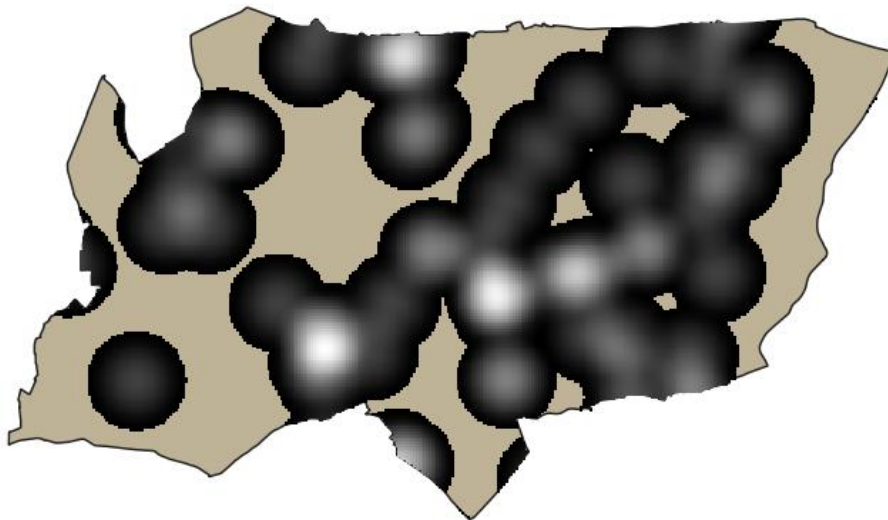
With the help of such maps, areas with crime ratio can be identified and such visualizations are essential in doing public safety analysis.

Session 8

Hotspot / Classification and Travel Cost / Accessibility (Spatial Clustering Analysis and Network Analysis with GeoDa and QGIS)

Task 1: Export the heatmap

Result:



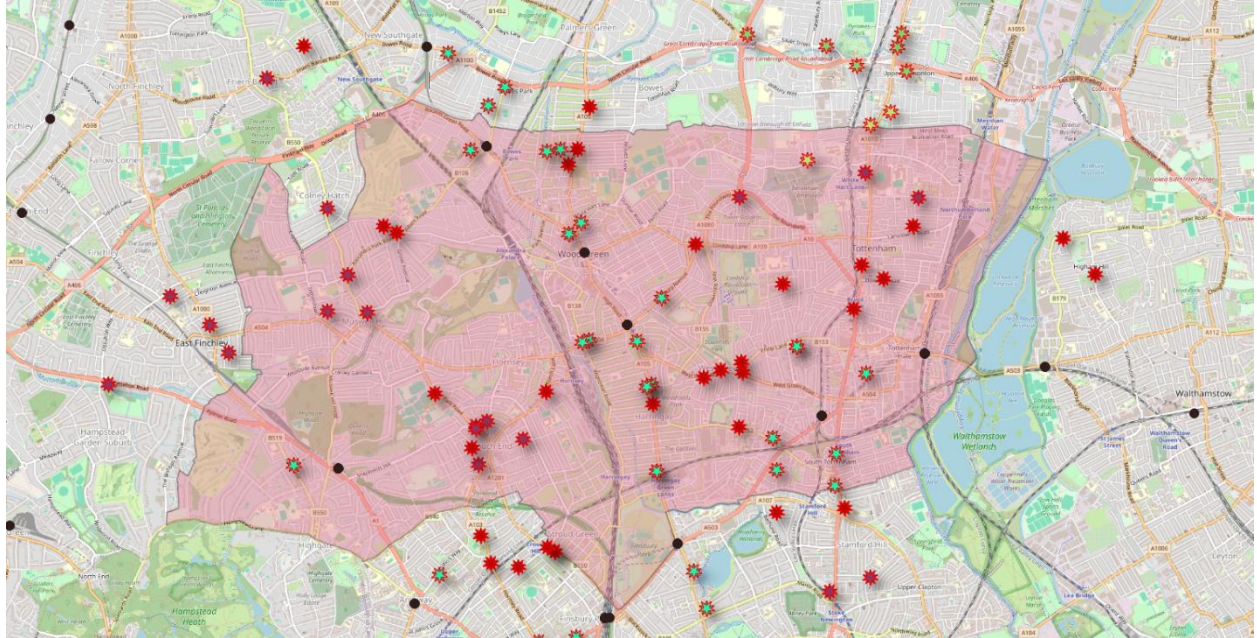
Description

This task depicts the use of clustering technique like Kernel Density Estimation (KDE) to identify service hotspots which in this case are GP surgeries, in one of the London boroughs (Haringey).

The spatial pattern reflects that some parts of Haringey have better access to different healthcare facilities, this is more likely because of higher population densities or the infrastructure of urban areas.

Task 4: The Accessibility map

Result:



Description

In this task, the accessibility of GP surgeries to the nearest tube stations in Haringey is measured with the help of QGIS and SpatiaLite. A distance matrix was created, and results were demonstrated or visualized through a graduated map. The results suggest that in central and eastern Haringey, GP surgeries have better accessibility to tube stations.

Applications of such accessibility maps include urban and health planning, creating policies for transport and allocation of resources.

Session 9

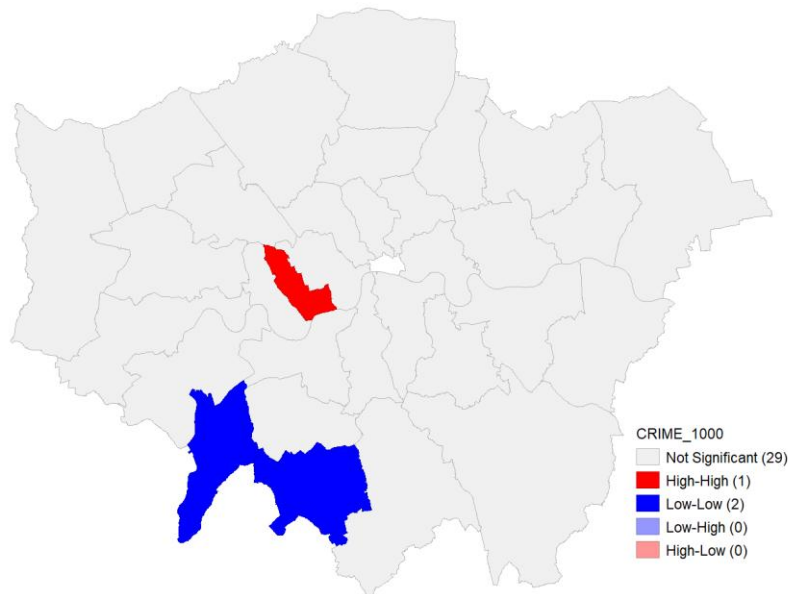
Near Things More Related Than Distance Things in London (Spatial Dependence, Spatial Regression and Surface Analysis with GeoDa and QGIS)

Task 1: Export specified Scatter Plots and Cluster Maps

Three variables for Univariate Local Moran's I

1) Crime_1000

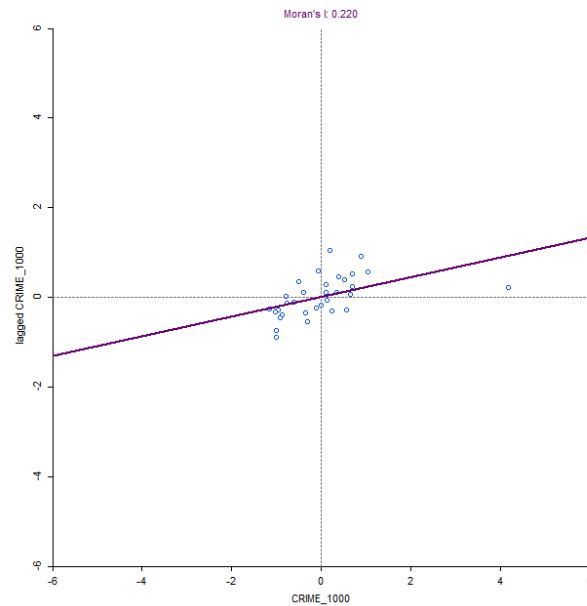
Cluster Map:



Description

Univariate Moran's I Cluster map of each borough with their spatial dependence on Crime_1000. One borough (Kensington and Chelsea) indicated in red shows high crime dependence while 2 boroughs (Kingston Upon Thames and Sutton) in blue show low crime dependence within these boroughs.

Scatter Plot:

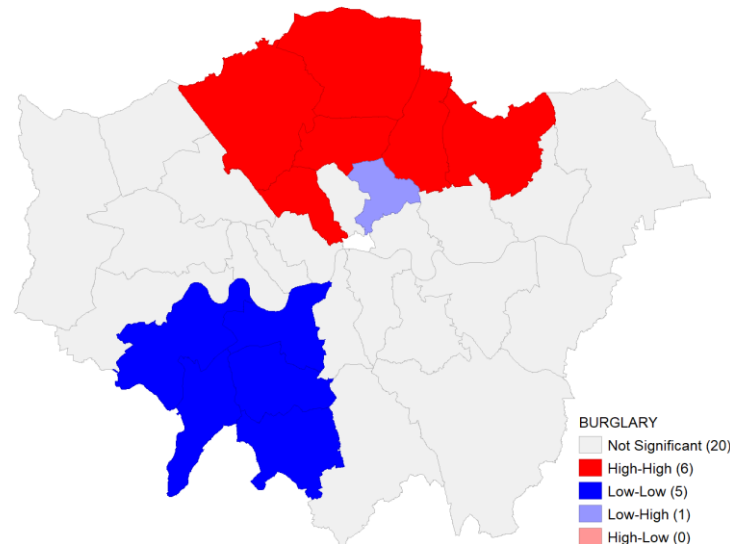


Description

The above graph depicts the **scatter plot** of Crimes_1000s using Univariate Moran's I .

2) Burglary

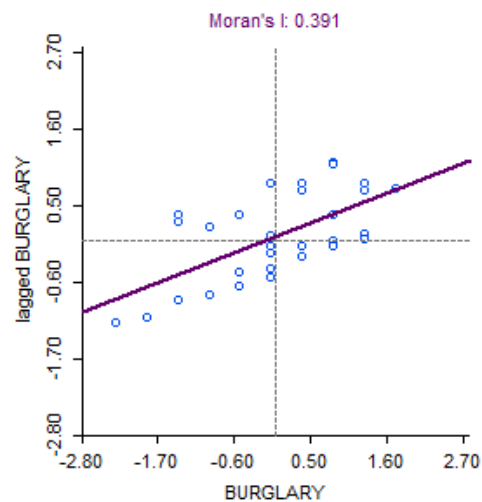
Cluster map:



Description

Univariate Moran's I Cluster map of Burglary. Hackney is the only borough in Inner London that shows a spatial correlation of Low-High. 5 southern boroughs indicate spatial autocorrelation of Low-Low while 6 Boroughs in the north represent High-High spatial autocorrelation. Other boroughs do not exhibit significant spatial correlations.

Scatter Plot:

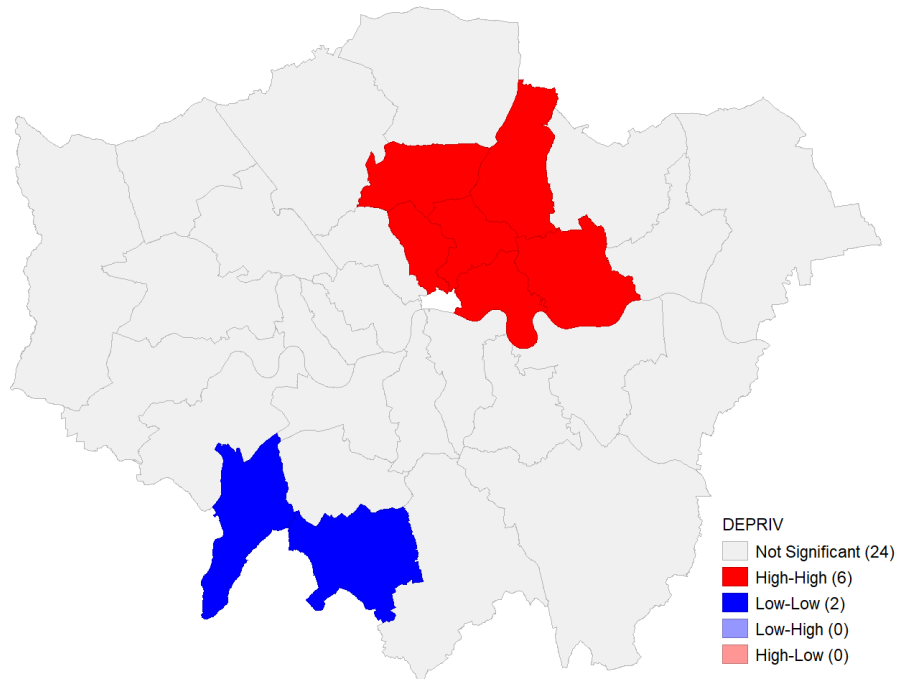


Description

The above graph is a Scatter Plot with a Univariate Moran's I of Burglary.

3) DEPRIV

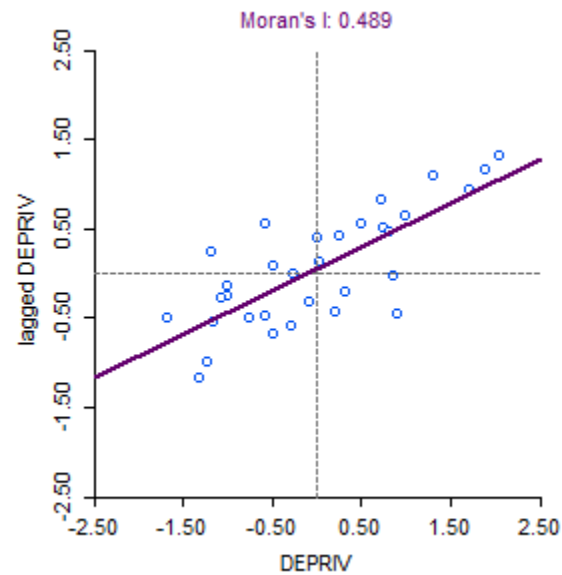
Cluster map:



Description

Cluster map of each borough with their spatial dependence on Deprivation. This map depicts that 2 southern boroughs indicated by blue show low deprivation while 6 boroughs in red represent high spatial autocorrelation.

Scatter Plot:



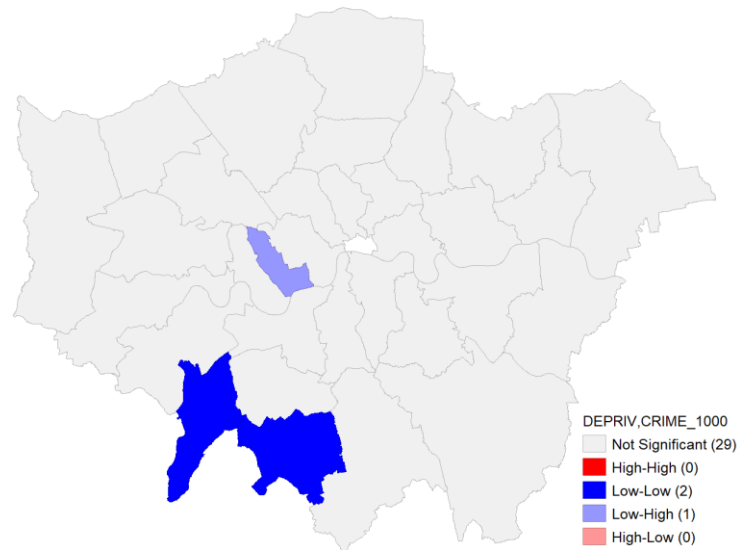
Description

The above figure shows the Moran's I plot of London borough with the spatial autocorrelation of their Deprivations.

One pair of variables for Bivariate Local Moran's I

Deprive, Crime_1000

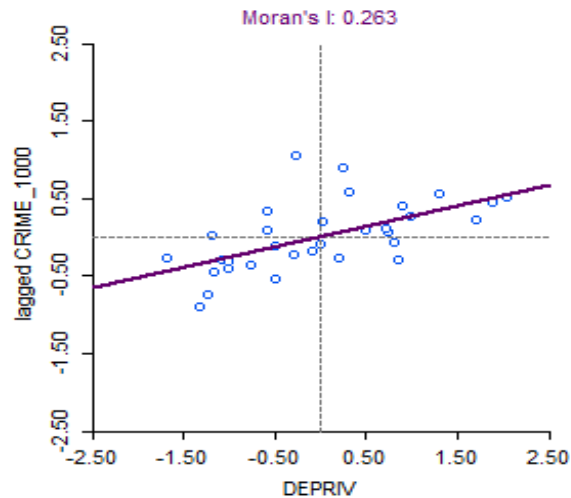
Cluster map:



Description

The above figure is a Bivariate Moran's I Cluster map of Deprivation against Crime_1000. Only one borough in Inner London shows a spatial correlation of Low-High and two boroughs show Low-Low spatial autocorrelation.

Scatter Plot:

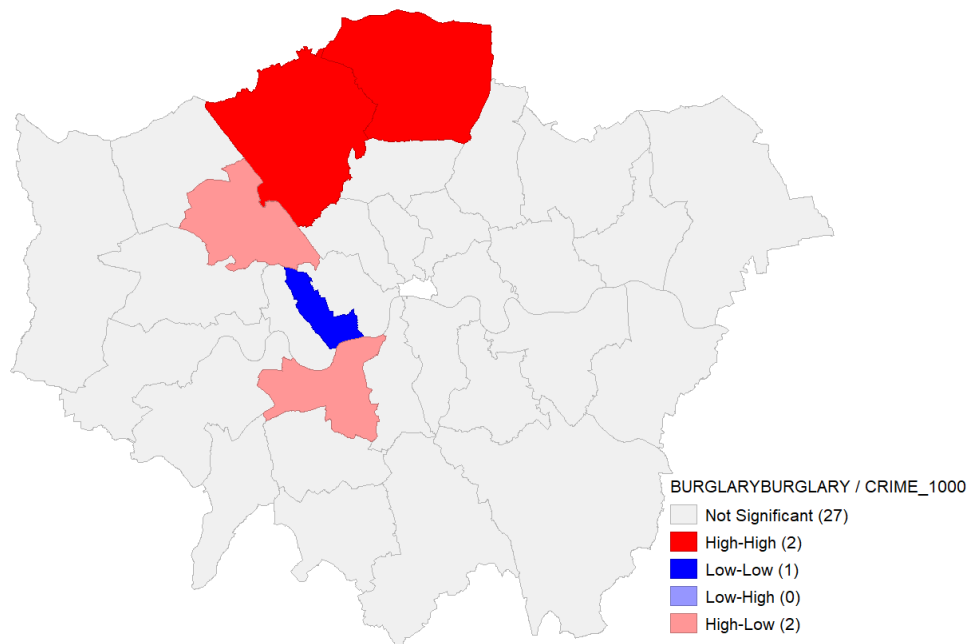


Description

The above figure is a Scatter Plot with a Bivariate Moran's I with Crime_1000 against Deprivation. It indicates a moderate but positive spatial autocorrelation.

One variable for Local Moran's I with EB Rate

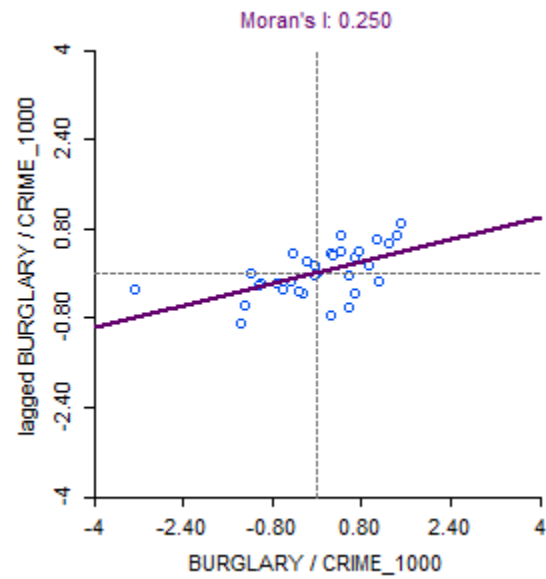
Cluster map:



Description

Cluster map of Moran's I with EB Rate I of Burglary/Crime_1000

Scatter Plot:

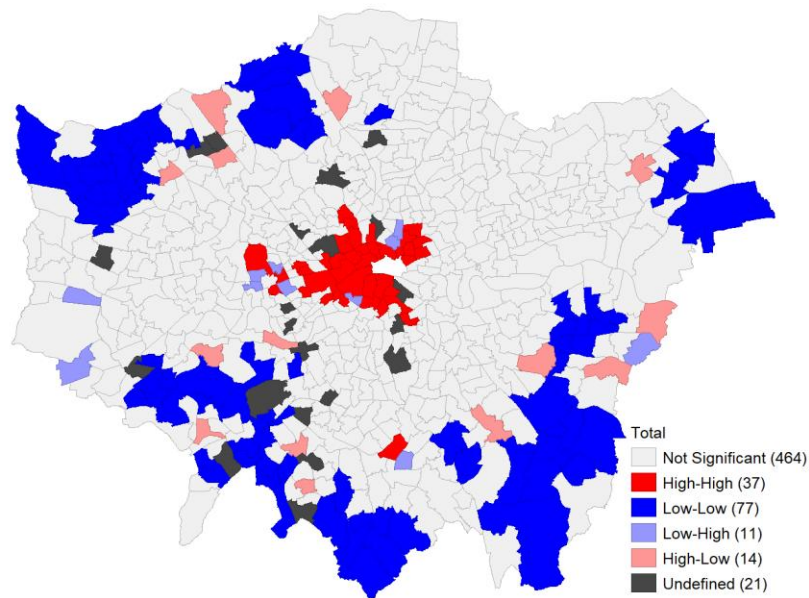


Description

Scatter Plot of Moran's I with EB Rate I of Burglary/Crime_1000.

One variable for “london ward met per1000.shp” with Univariate Local Moran’s I

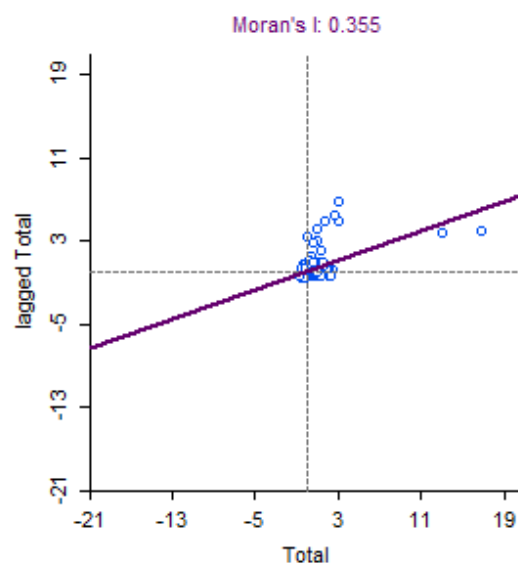
Cluster map:



Description

This is a Univariate Local Moran's I Cluster map of Total Crime/1000 in London.

Scatter Plot:

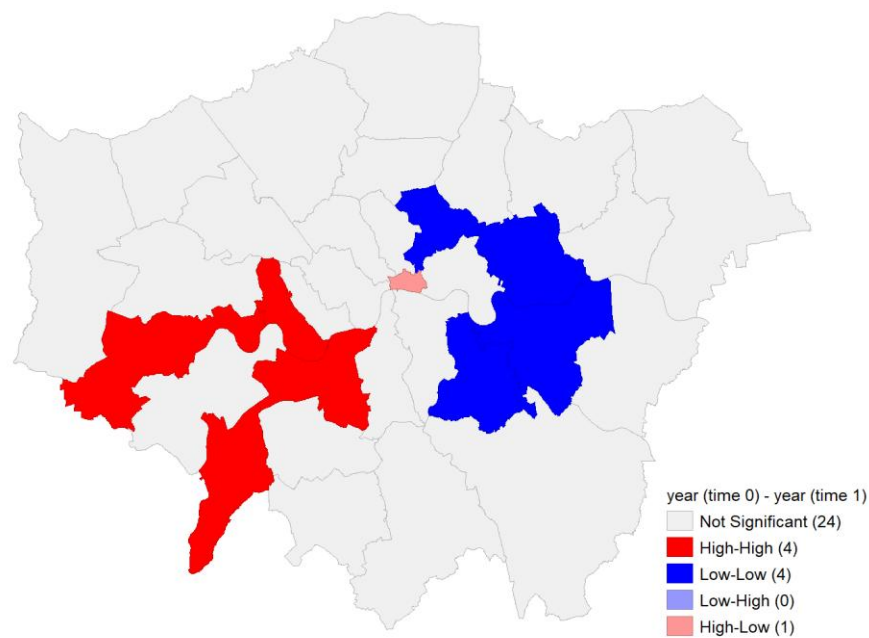


Description

This is a Scatter Plot of Univariate Local Moran's I . The plot depicts Moran's I of the total counts of crime per thousand people in London. Rock Contiguity was utilized as the Weight Manager in GeoDa.

The population estimates in 2011 and 2016 for Differential Local Moran's I

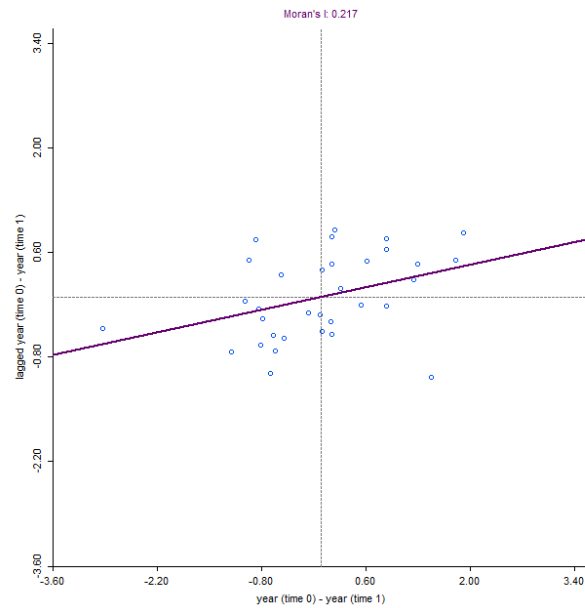
Cluster map:



Description

The above map is a Differential Local Moran's Cluster map of spatial-temporal dependence by population estimates from 2011 and 2016 in London. This map indicates that there was a significant spatiotemporal difference between the two-time frames

Scatter Plot:



Description

The above figure is a Scatter Plot of Differential Local Moran's I. This scatter plot shows spatial-temporal dependence by population estimates between 2011 and 2016.

Task 2: Copy results of regression modelling

Result (Spatial lag):

>>04/28/25 18:42:31

REGRESSION

SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION

Data set : london_life_polygon

Spatial Weight : london_life_polygon

Dependent Variable : LIFE_MALE Number of Observations: 32

Mean dependent var : 77.4125 Number of Variables : 7

S.D. dependent var : 1.8562 Degrees of Freedom : 25

Lag coeff. (Rho) : -0.0139335

R-squared : 0.803713 Log likelihood : -39.1488

Sq. Correlation : - Akaike info criterion : 92.2976

Sigma-square : 0.6763 Schwarz criterion : 102.558

S.E of regression : 0.822375

Variable	Coefficient	Std.Error	z-value	Probability
W_LIFE_MALE	-0.0139335	0.180354	-0.0772563	0.93842
CONSTANT	70.3696	13.9556	5.04239	0.00000
DEPRIV	-0.0412571	0.0328003	-1.25783	0.20845
P_SMOKE	0.100128	0.0760677	1.3163	0.18807
P_BINGE	-0.355534	0.203054	-1.75093	0.07996

P_OBESE	0.0469733	0.0834921	0.562607	0.57370
P_HEALTHY	0.353448	0.0717953	4.923	0.00000

REGRESSION DIAGNOSTICS

DIAGNOSTICS FOR HETEROSKEDASTICITY

RANDOM COEFFICIENTS

TEST	DF	VALUE	PROB
Breusch-Pagan test	5	5.7305	0.33333

DIAGNOSTICS FOR SPATIAL DEPENDENCE

SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX : london_life_polygon

TEST	DF	VALUE	PROB
Likelihood Ratio Test	1	0.0053	0.94215

Description

SPATIAL LAG regression model of health variables from GeoDa.

Results (Classis):

>>04/28/25 18:38:19

REGRESSION

SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES ESTIMATION

Data set : london_life_polygon

Dependent Variable : LIFE_MALE Number of Observations: 32

Mean dependent var : 77.4125 Number of Variables : 6

S.D. dependent var : 1.8562 Degrees of Freedom : 26

R-squared : 0.803673 F-statistic : 21.2864
 Adjusted R-squared : 0.765918 Prob(F-statistic) : 1.92573e-08
 Sum squared residual: 21.6461 Log likelihood : -39.1514
 Sigma-square : 0.832541 Akaike info criterion : 90.3029
 S.E. of regression : 0.912437 Schwarz criterion : 99.0973
 Sigma-square ML : 0.676439
 S.E of regression ML: 0.822459

Variable	Coefficient	Std.Error	t-Statistic	Probability
CONSTANT	69.3676	3.95948	17.5194	0.00000
DEPRIV	-0.0399811	0.0302057	-1.32363	0.19715
P_SMOKE	0.101164	0.0802025	1.26135	0.21838
P_OBESE	0.0444868	0.0876369	0.507626	0.61599
P_BINGE	-0.361891	0.198456	-1.82353	0.07974
P_HEALTHY	0.35326	0.0791595	4.46264	0.00014

REGRESSION DIAGNOSTICS

MULTICOLLINEARITY CONDITION NUMBER 70.463146

TEST ON NORMALITY OF ERRORS

TEST	DF	VALUE	PROB
Jarque-Bera	2	1.3617	0.50619

DIAGNOSTICS FOR HETEROSKEDASTICITY

RANDOM COEFFICIENTS

TEST	DF	VALUE	PROB
Breusch-Pagan test	5	5.7773	0.32849

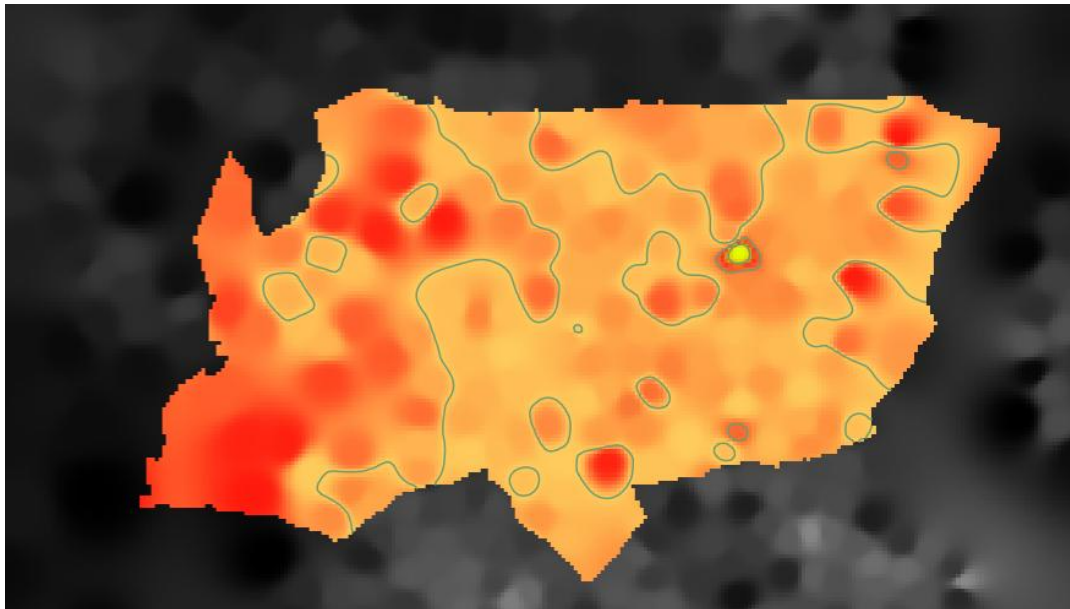
Koenker-Bassett test 5 6.3910 0.27001

Description

Classic regression model of health variables from GeoDa.

Task 3: Export the interpolated raster image

Result:



Description

In this task, the population density surface is created for Haringey using IDW interpolation, distance coefficient is set to value 4 and clipped to borough boundaries. Then added 100-person contour lines.

Such mappings are crucial for urban planning and can be used to identify high population areas for better resource allocation.